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$$\int x\sqrt{1-4x^2}dx$$

$$\text{Stel } t = 1 - 4x^2 \Rightarrow dt = -8xdx \Rightarrow -\frac{1}{8}dt = dx$$

$$= -\frac{1}{8} \int \sqrt{t}dt$$

$$= -\frac{2 \cdot \sqrt{1-4x^2} \cdot (1-4x^2)}{24} + C$$

$$= -\frac{\sqrt{1-4x^2} \cdot (1-4x^2)}{12} + C$$

References